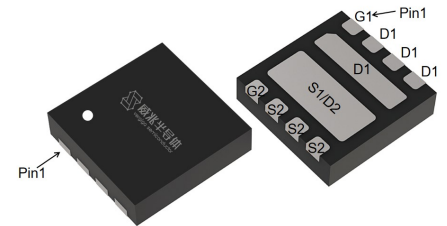


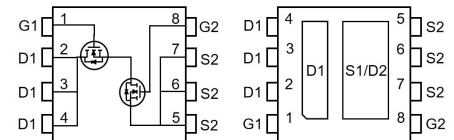
## Features

- Dual Asymmetric N-Channel
- VitoMOS® II Technology
- 100% Avalanche Tested, 100% Rg Tested

|                                 |     |     |    |
|---------------------------------|-----|-----|----|
| $V_{DS}$                        | 30  | 30  | V  |
| $R_{DS(on),TYP@ V_{GS}=10V}$    | 8.2 | 6.5 | mΩ |
| $R_{DS(on),TYP@ V_{GS}=4.5V}$   | 12  | 11  | mΩ |
| $I_D(\text{Wire bond Limited})$ | 24  | 30  | A  |


**DFN3x3**


| Part ID  | Package Type | Marking | Packing      |
|----------|--------------|---------|--------------|
| VS3652DB | DFN3x3       | 3652DB  | 5000pcs/Reel |



## Maximum ratings, at $T_A = 25^\circ\text{C}$ , unless otherwise specified

| Symbol         | Parameter                                                   |                           | Rating     |            | Unit             |
|----------------|-------------------------------------------------------------|---------------------------|------------|------------|------------------|
|                |                                                             |                           | Q1         | Q2         |                  |
| $V_{(BR)DSS}$  | Drain-Source breakdown voltage                              |                           | 30         | 30         | V                |
| $V_{GS}$       | Gate-Source voltage                                         |                           | $\pm 20$   | $\pm 20$   | V                |
| $I_S$          | Diode continuous forward current                            | $T_C = 25^\circ\text{C}$  | 8.3        | 9.2        | A                |
| $I_D$          | Continuous drain current @ $V_{GS}=10V$ (Wire bond Limited) | $T_C = 25^\circ\text{C}$  | 24         | 30         | A                |
|                | Continuous drain current @ $V_{GS}=10V$ (Silicon Limited)   | $T_C = 100^\circ\text{C}$ | 16         | 19         | A                |
| $I_{DM}$       | Pulse drain current tested ①                                | $T_C = 25^\circ\text{C}$  | 100        | 120        | A                |
| $I_{DSM}$      | Continuous drain current @ $V_{GS}=10V$                     | $T_A = 25^\circ\text{C}$  | 10         | 11         | A                |
|                |                                                             | $T_A = 70^\circ\text{C}$  | 8          | 9          | A                |
| $E_{AS}$       | Avalanche energy, single pulsed ②                           |                           | 49         | 49         | mJ               |
| $P_D$          | Maximum power dissipation ③                                 | $T_C = 25^\circ\text{C}$  | 10         | 11         | W                |
| $P_{DSM}$      | Maximum power dissipation ④                                 | $T_A = 25^\circ\text{C}$  | 1.5        | 1.6        | W                |
| $T_{STG}, T_J$ | Storage and Junction Temperature Range                      |                           | -55 to 150 | -55 to 150 | $^\circ\text{C}$ |

## Thermal Characteristics

| Symbol          | Parameter                                 | Typical |    | Max |    | Unit               |
|-----------------|-------------------------------------------|---------|----|-----|----|--------------------|
|                 |                                           | Q1      | Q2 | Q1  | Q2 |                    |
| $R_{\theta JC}$ | Thermal Resistance, Junction-to-Case ⑤    | 10      | 9  | 12  | 11 | $^\circ\text{C/W}$ |
| $R_{\theta JA}$ | Thermal Resistance, Junction-to-Ambient ⑥ | 70      | 65 | 84  | 78 | $^\circ\text{C/W}$ |

**Q1-Channel Electrical Characteristics**

| Symbol                                                                               | Parameter                                               | Condition                                                                                    | Min | Typ | Max  | Unit |
|--------------------------------------------------------------------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------|-----|-----|------|------|
| Static Electrical Characteristics @ T <sub>J</sub> = 25°C (unless otherwise stated)  |                                                         |                                                                                              |     |     |      |      |
| V(BR)DSS                                                                             | Drain-Source Breakdown Voltage                          | V <sub>GS</sub> =0V, I <sub>D</sub> =250μA                                                   | 30  | --  | --   | V    |
| I <sub>DSS</sub>                                                                     | Zero Gate Voltage Drain Current(T <sub>J</sub> =25°C)   | V <sub>DS</sub> =30V,V <sub>GS</sub> =0V                                                     | --  | --  | 1    | μA   |
|                                                                                      | Zero Gate Voltage Drain Current(T <sub>J</sub> =125°C)⑦ | V <sub>DS</sub> =30V,V <sub>GS</sub> =0V                                                     | --  | --  | 100  | μA   |
| I <sub>GSS</sub>                                                                     | Gate-Body Leakage Current                               | V <sub>GS</sub> =±20V,V <sub>DS</sub> =0V                                                    | --  | --  | ±100 | nA   |
| V <sub>GS(th)</sub>                                                                  | Gate Threshold Voltage                                  | V <sub>DS</sub> =V <sub>GS</sub> ,I <sub>D</sub> =250μA                                      | 1.2 | 1.7 | 2.3  | V    |
| R <sub>DS(on)</sub>                                                                  | Drain-Source On-State Resistance ⑧                      | V <sub>GS</sub> =10V, I <sub>D</sub> =13A                                                    | --  | 8.2 | 11   | mΩ   |
|                                                                                      |                                                         | V <sub>GS</sub> =4.5V, I <sub>D</sub> =10A                                                   | --  | 12  | 16   | mΩ   |
| Dynamic Electrical Characteristics @ T <sub>J</sub> = 25°C (unless otherwise stated) |                                                         |                                                                                              |     |     |      |      |
| C <sub>iss</sub>                                                                     | Input Capacitance ⑦                                     | V <sub>DS</sub> =15V,V <sub>GS</sub> =0V,<br>f=1MHz                                          | --  | 420 | --   | pF   |
| C <sub>oss</sub>                                                                     | Output Capacitance ⑦                                    |                                                                                              | --  | 315 | --   | pF   |
| C <sub>rss</sub>                                                                     | Reverse Transfer Capacitance ⑦                          |                                                                                              | --  | 10  | --   | pF   |
| R <sub>g</sub>                                                                       | Gate Resistance                                         | f=1MHz                                                                                       | --  | 1.3 | --   | Ω    |
| Q <sub>g</sub> (10V)                                                                 | Total Gate Charge ⑦                                     | V <sub>DS</sub> =15V,I <sub>D</sub> =13A,<br>V <sub>GS</sub> =10V                            | --  | 7.5 | --   | nC   |
| Q <sub>g</sub> (4.5V)                                                                | Total Gate Charge ⑦                                     |                                                                                              | --  | 3.6 | --   | nC   |
| Q <sub>gs</sub>                                                                      | Gate Source Charge ⑦                                    |                                                                                              | --  | 1.6 | --   | nC   |
| Q <sub>gd</sub>                                                                      | Gate Drain Charge ⑦                                     |                                                                                              | --  | 1.1 | --   | nC   |
| Switching Characteristics ⑦                                                          |                                                         |                                                                                              |     |     |      |      |
| T <sub>d(on)</sub>                                                                   | Turn on Delay Time                                      | V <sub>DD</sub> =15V,<br>I <sub>D</sub> =13A,<br>R <sub>G</sub> =3Ω,<br>V <sub>GS</sub> =10V | --  | 5.6 | --   | ns   |
| T <sub>r</sub>                                                                       | Turn on Rise Time                                       |                                                                                              | --  | 48  | --   | ns   |
| T <sub>d(off)</sub>                                                                  | Turn Off Delay Time                                     |                                                                                              | -   | 12  | --   | ns   |
| T <sub>f</sub>                                                                       | Turn Off Fall Time                                      |                                                                                              | --  | 2.6 | --   | ns   |
| Source Drain Diode Characteristics                                                   |                                                         |                                                                                              |     |     |      |      |
| V <sub>SD</sub>                                                                      | Forward on voltage                                      | I <sub>SD</sub> =10A,V <sub>GS</sub> =0V                                                     | --  | 0.8 | 1.2  | V    |
| T <sub>rr</sub>                                                                      | Reverse Recovery Time ⑦                                 | I <sub>sd</sub> =13A, V <sub>GS</sub> =0V                                                    | --  | 13  | --   | ns   |
| Q <sub>rr</sub>                                                                      | Reverse Recovery Charge ⑦                               | di/dt=100A/μs                                                                                | --  | 3.5 | --   | nC   |

NOTE:

① Single pulse; pulse width ≤ 100μs.

② Q1: EAS of 49mJ is based on starting T<sub>J</sub> = 25°C, L = 0.5mH, R<sub>G</sub> = 25Ω, I<sub>AS</sub> = 14A, V<sub>GS</sub>=10V; 100% FT tested at L = 0.5mH, I<sub>AS</sub> = 8A.

Q2: EAS of 49mJ is based on starting T<sub>J</sub> = 25°C, L = 0.5mH, R<sub>G</sub> = 25Ω, I<sub>AS</sub> = 14A, V<sub>GS</sub>=10V; 100% FT tested at L = 0.5mH, I<sub>AS</sub> = 8A.

③ The power dissipation P<sub>d</sub> is based on T<sub>J</sub>(max), using junction-to-case thermal resistance RθJC.

④ The power dissipation P<sub>dsm</sub> is based on T<sub>J</sub>(max), using junction-to-ambient thermal resistance RθJA.

⑤ These tests are performed respectively with the device mounted on a 1 in2 pad and a minimum pad of 2oz. Copper FR-4 board in a still air environment with TA=25°C, using Transient Dual Interface method to acquire RθJC.

⑥ These tests are performed with the device mounted on 1 in2 FR-4 board with 2oz. Copper, in a still air environment with TA=25°C.

⑦ Guaranteed by design, not subject to production testing.

⑧ Pulse width ≤ 380μs; duty cycles ≤ 2%.

## Q2-Channel Electrical Characteristics

| Symbol                                                                               | Parameter                                                | Condition                                               | Min | Typ | Max  | Unit |
|--------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------------------|-----|-----|------|------|
| Static Electrical Characteristics @ T <sub>J</sub> = 25°C (unless otherwise stated)  |                                                          |                                                         |     |     |      |      |
| V(BR)DSS                                                                             | Drain-Source Breakdown Voltage                           | V <sub>GS</sub> =0V, I <sub>D</sub> =250μA              | 30  | --  | --   | V    |
| IDSS                                                                                 | Zero Gate Voltage Drain Current t(T <sub>J</sub> =25°C)  | V <sub>DS</sub> =30V,V <sub>GS</sub> =0V                | --  | --  | 1    | μA   |
|                                                                                      | Zero Gate Voltage Drain Current t(T <sub>J</sub> =125°C) | V <sub>DS</sub> =30V,V <sub>GS</sub> =0V                | --  | --  | 100  | μA   |
| IGSS                                                                                 | Gate-Body Leakage Current                                | V <sub>GS</sub> =±20V,V <sub>DS</sub> =0V               | --  | --  | ±100 | nA   |
| VGS(th)                                                                              | Gate Threshold Voltage                                   | V <sub>DS</sub> =V <sub>GS</sub> ,I <sub>D</sub> =250μA | 1.2 | 1.7 | 2.3  | V    |
| RDS(on)                                                                              | Drain-Source On-State Resistance ④                       | V <sub>GS</sub> =10V, I <sub>D</sub> =15A               | --  | 6.5 | 8.5  | mΩ   |
|                                                                                      |                                                          | V <sub>GS</sub> =4.5V, I <sub>D</sub> =10A              | --  | 11  | 14   | mΩ   |
| Dynamic Electrical Characteristics @ T <sub>J</sub> = 25°C (unless otherwise stated) |                                                          |                                                         |     |     |      |      |
| Ciss                                                                                 | Input Capacitance                                        | VDS=15V,VGS=0V,<br>f=1MHz                               | --  | 625 | --   | pF   |
| Coss                                                                                 | Output Capacitance                                       |                                                         | --  | 405 | --   | pF   |
| Crss                                                                                 | Reverse Transfer Capacitance                             |                                                         | --  | 45  | --   | pF   |
| Rg                                                                                   | Gate Resistance                                          | f=1MHz                                                  | --  | 1.9 | --   | Ω    |
| Qg(10V)                                                                              | Total Gate Charge                                        | VDS=15V,ID=15A,<br>VGS=10V                              | --  | 12  | --   | nC   |
| Qg(4.5V)                                                                             | Total Gate Charge                                        |                                                         | --  | 5.8 | --   | nC   |
| Qgs                                                                                  | Gate Source Charge                                       |                                                         | --  | 2.1 | --   | nC   |
| Qgd                                                                                  | Gate Drain Charge                                        |                                                         | --  | 2.3 | --   | nC   |
| Switching Characteristics                                                            |                                                          |                                                         |     |     |      |      |
| Td(on)                                                                               | Turn on Delay Time                                       | VDD=15V,<br>ID=15A,<br>RG=3Ω,<br>VGS=10V                | --  | 6.8 | --   | ns   |
| Tr                                                                                   | Turn on Rise Time                                        |                                                         | --  | 62  | --   | ns   |
| Td(off)                                                                              | Turn Off Delay Time                                      |                                                         | -   | 16  | --   | ns   |
| Tf                                                                                   | Turn Off Fall Time                                       |                                                         | --  | 29  | --   | ns   |
| Source Drain Diode Characteristics                                                   |                                                          |                                                         |     |     |      |      |
| VSD                                                                                  | Forward on voltage                                       | I <sub>SD</sub> =10A,V <sub>GS</sub> =0V                | --  | 0.8 | 1.2  | V    |
| Trr                                                                                  | Reverse Recovery Time                                    | I <sub>SD</sub> =15A, VGS=0V                            | --  | 18  | --   | ns   |
| Qrr                                                                                  | Reverse Recovery Charge                                  | di/dt=100A/μs                                           | --  | 6.0 | --   | nC   |

NOTE:

① Single pulse; pulse width ≤ 100μs.

② Q1: EAS of 49mJ is based on starting T<sub>J</sub> = 25°C, L = 0.5mH, R<sub>G</sub> = 25Ω, I<sub>AS</sub> = 14A, V<sub>GS</sub> = 10V; 100% FT tested at L = 0.5mH, I<sub>AS</sub> = 8A.

Q2: EAS of 49mJ is based on starting T<sub>J</sub> = 25°C, L = 0.5mH, R<sub>G</sub> = 25Ω, I<sub>AS</sub> = 14A, V<sub>GS</sub> = 10V; 100% FT tested at L = 0.5mH, I<sub>AS</sub> = 8A.

③ The power dissipation Pd is based on T<sub>J</sub>(max), using junction-to-case thermal resistance RθJC.

④ The power dissipation Pdsm is based on T<sub>J</sub>(max), using junction-to-ambient thermal resistance RθJA.

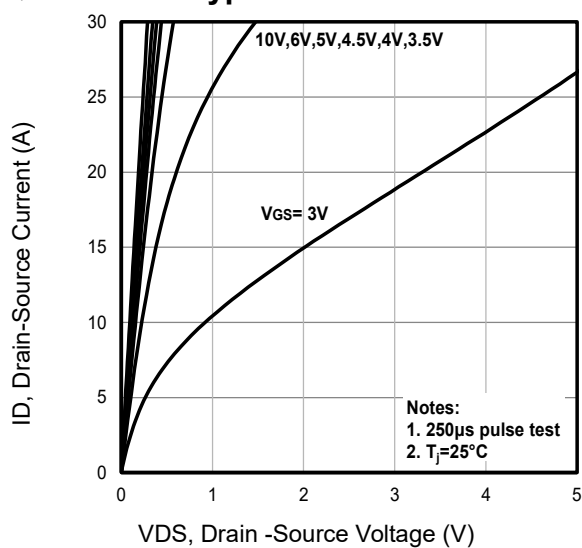
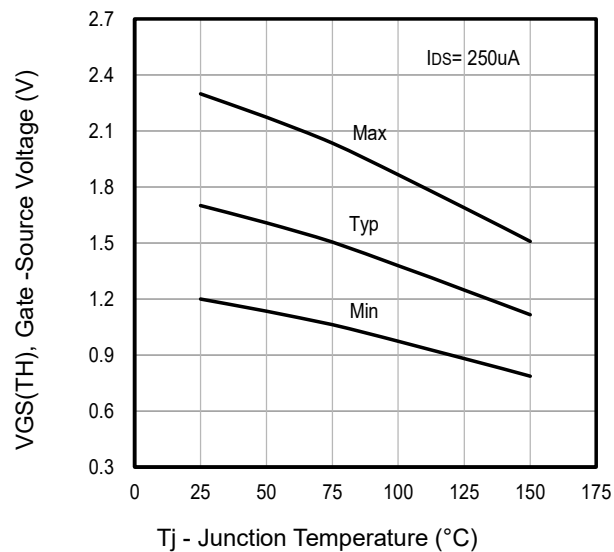
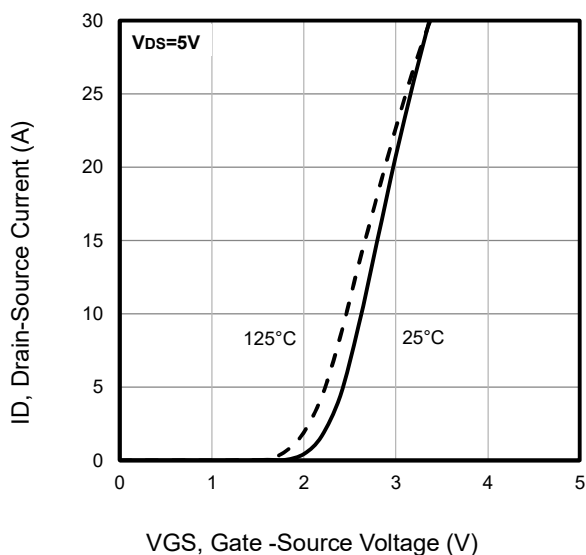
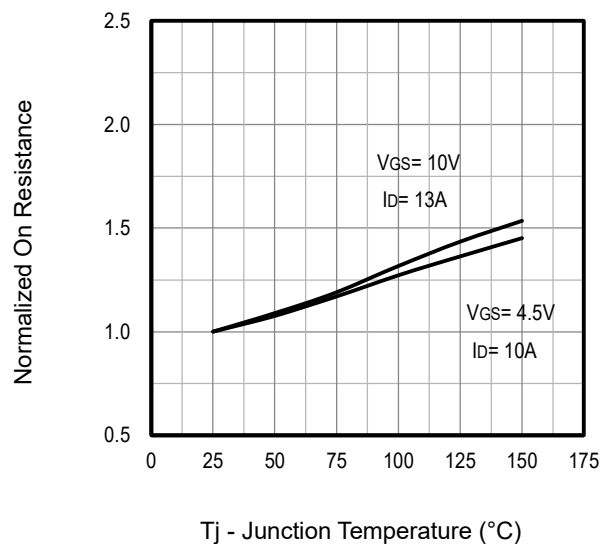
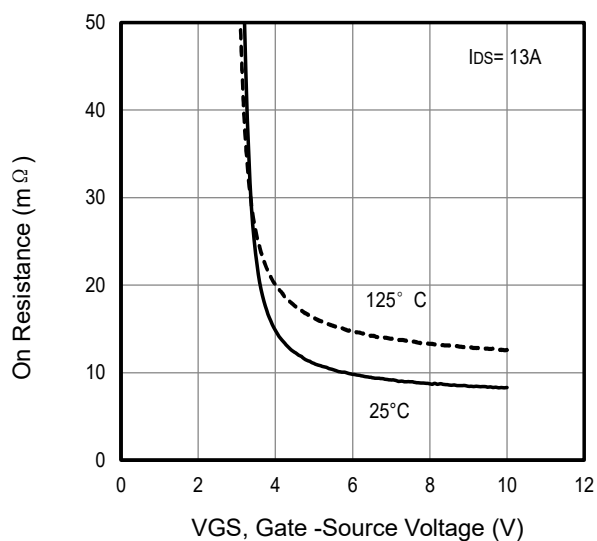
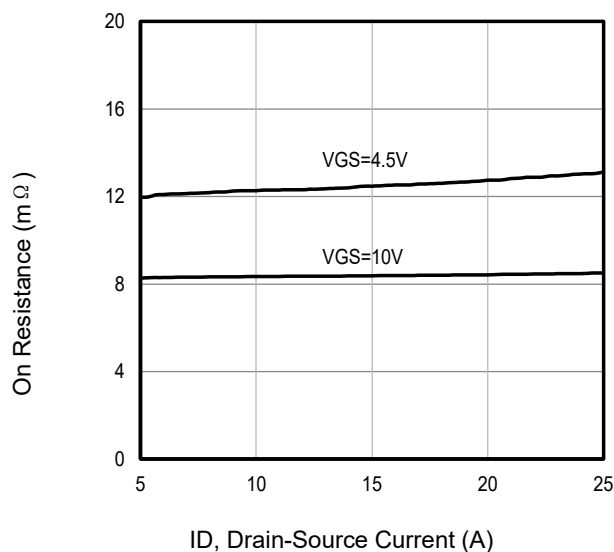
⑤ These tests are performed respectively with the device mounted on a 1 in2 pad and a minimum pad of 2oz. Copper FR-4 board in a still air environment with TA=25°C, using Transient Dual Interface method to acquire RθJC.

⑥ These tests are performed with the device mounted on 1 in2 FR-4 board with 2oz. Copper, in a still air environment with TA=25°C.

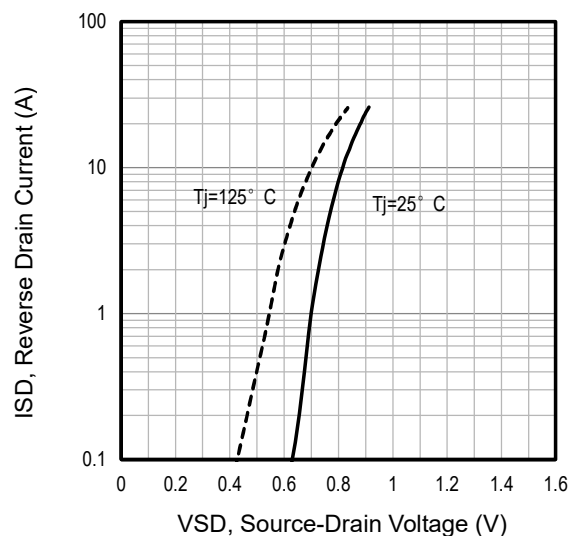
⑦ Guaranteed by design, not subject to production testing.

⑧ Pulse width ≤ 380μs; duty cycles ≤ 2%.

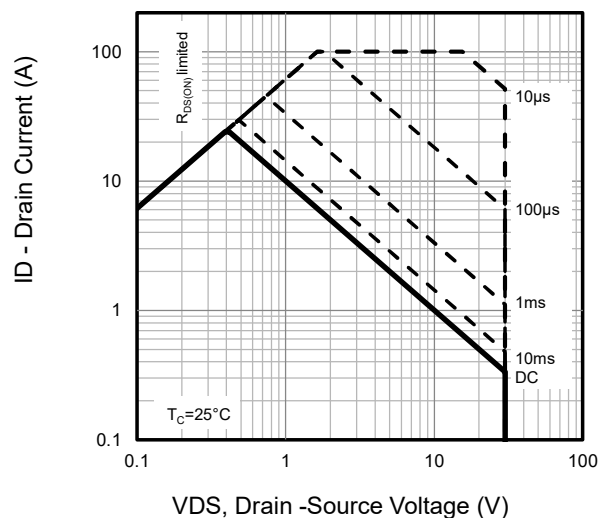
## Q1-Channel Typical Characteristics


**Fig1.** Typical Output Characteristics

**Fig2.** Typical  $V_{GS(TH)}$  Gate-Source Voltage Vs.  $T_j$ 

**Fig3.** Typical Transfer Characteristics

**Fig4.** Typical Normalized On-Resistance Vs.  $T_j$ 

**Fig5.** Typical On Resistance Vs Gate-Source Voltage

**Fig6.** Typical On Resistance Vs Drain Current

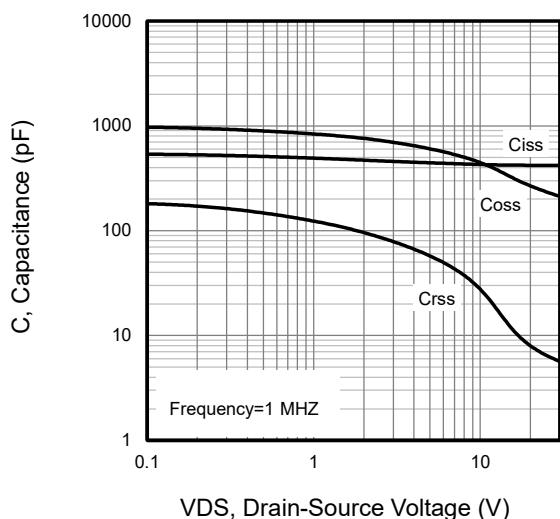
## Q1-Channel Typical Characteristics



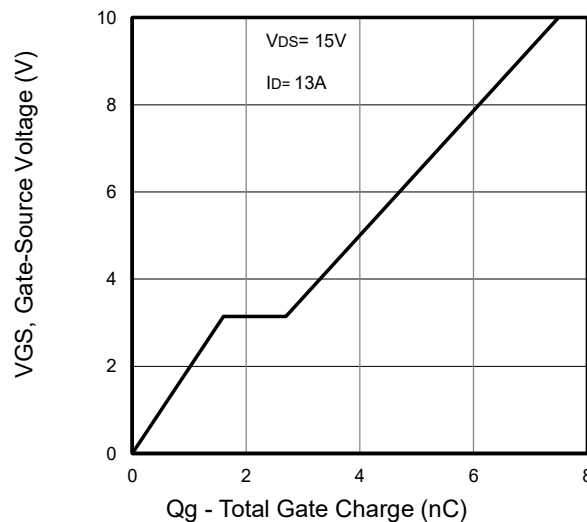
**Fig7.** Typical Source-Drain Diode Forward Voltage



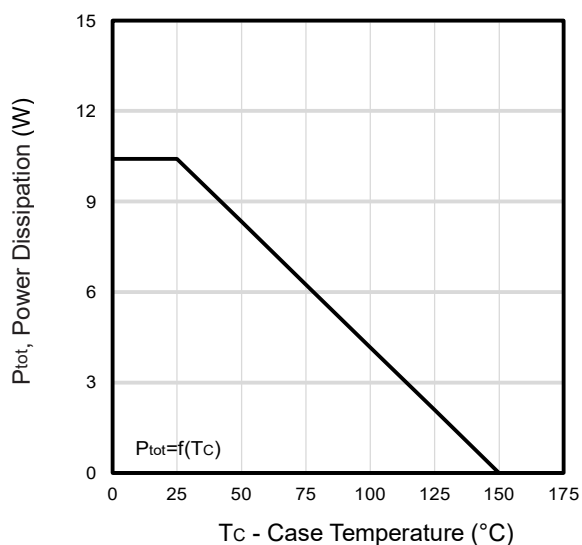
**Fig8.** Maximum Safe Operating Area



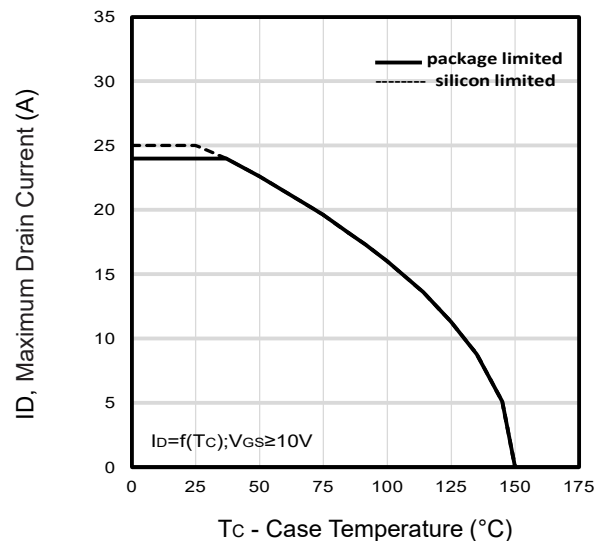
**Fig9.** Typical Capacitance Vs. Drain-Source Voltage



**Fig10.** Typical Gate Charge Vs. Gate-Source Voltage

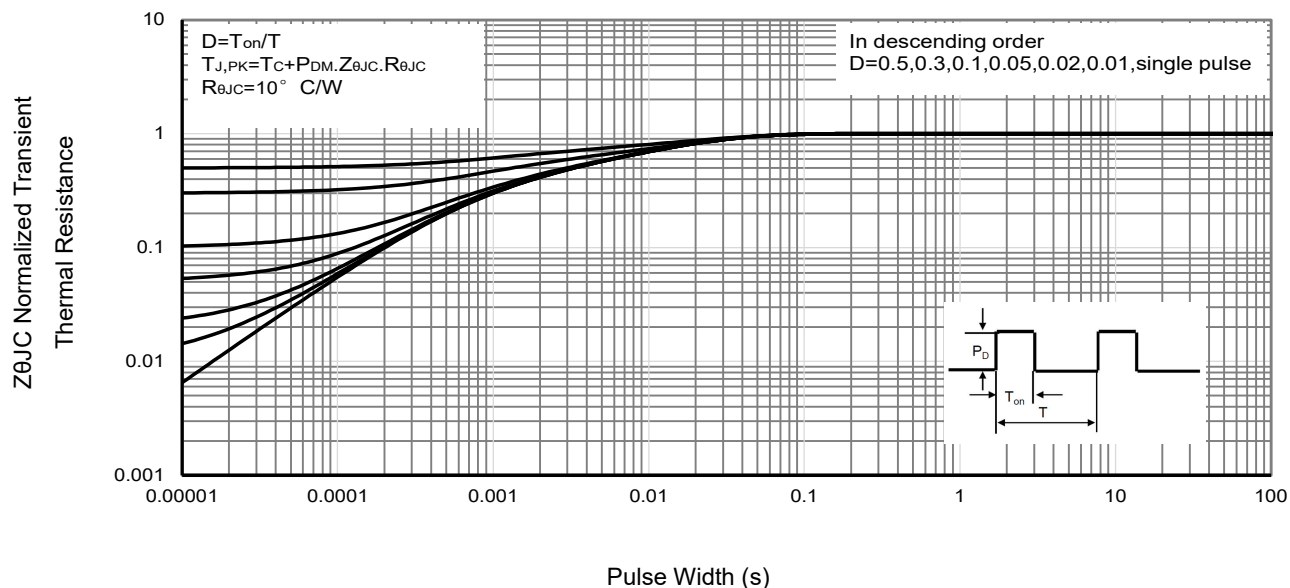


**Fig11.** Power Dissipation Vs. Case Temperature

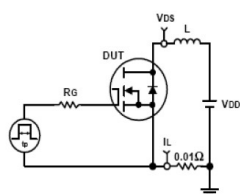


**Fig12.** Maximum Drain Current Vs. Case Temperature

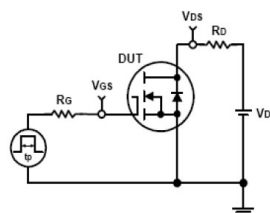
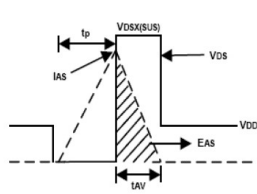
## Q1-Channel Typical Characteristics



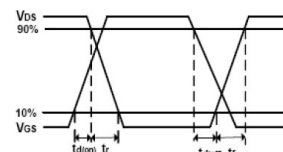
**Fig13 . Normalized Maximum Transient Thermal Impedance**



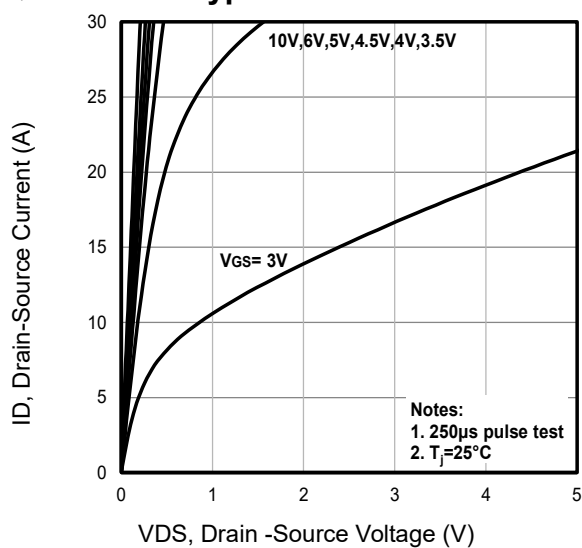
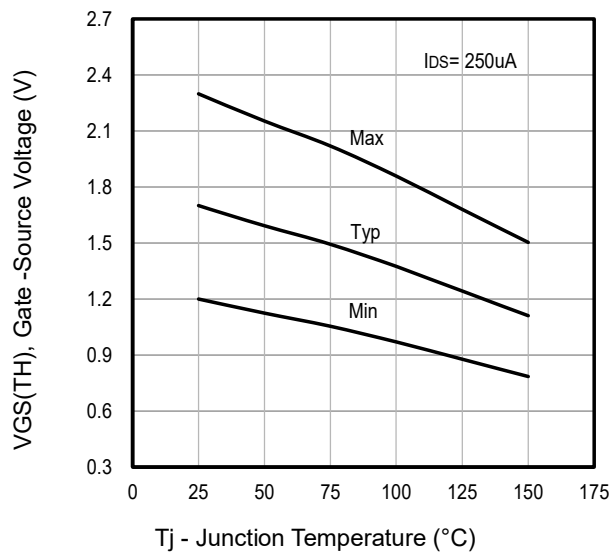
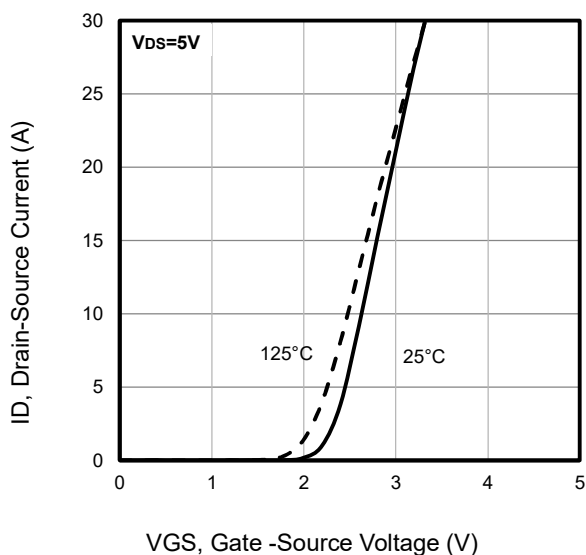
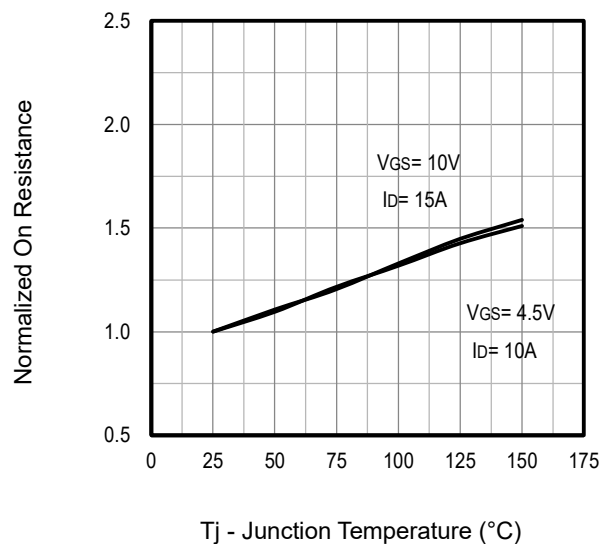
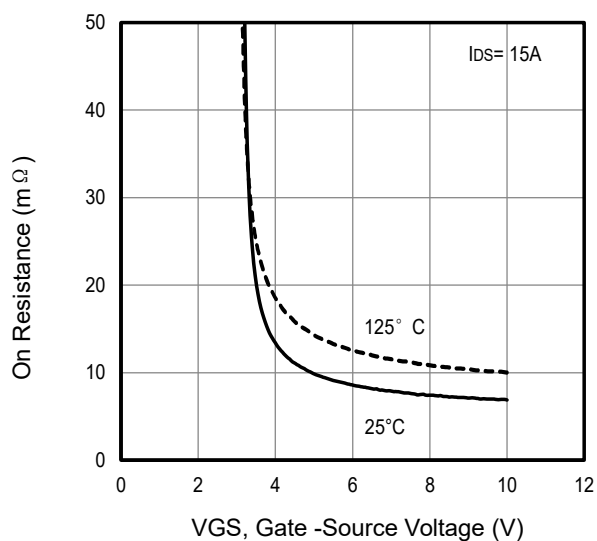
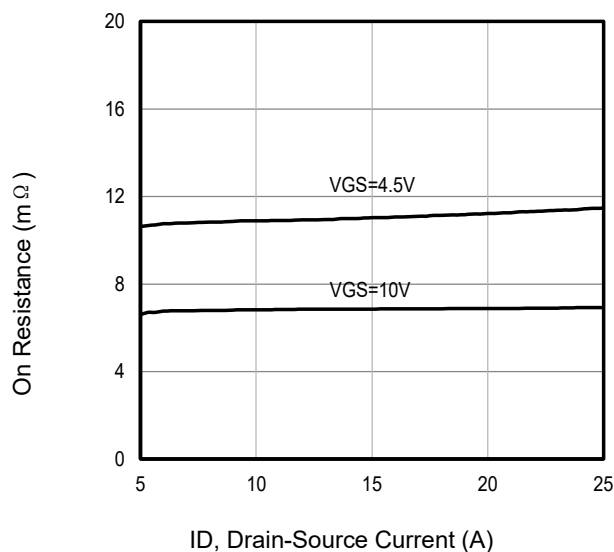
**Fig14. Unclamped Inductive Test Circuit and waveforms**



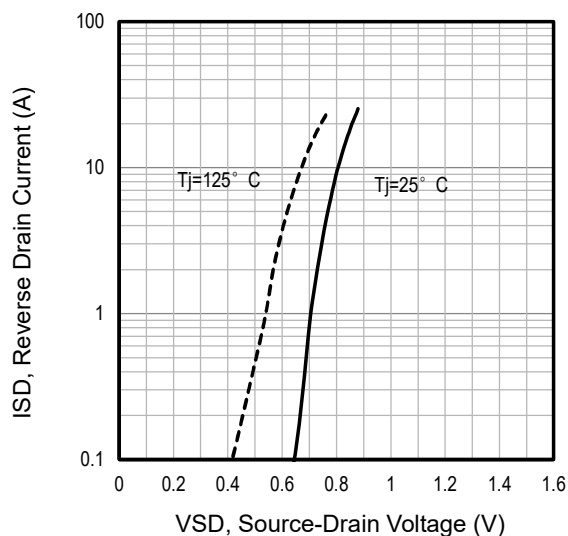
**Fig15. Switching Time Test Circuit and waveforms**



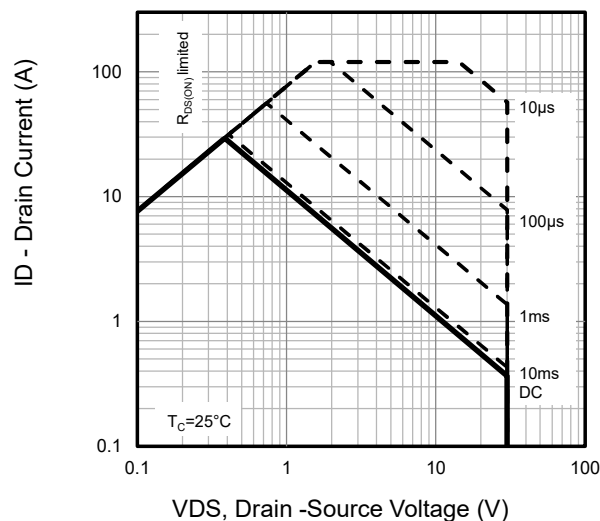
## Q2-Channel Typical Characteristics


**Fig1.** Typical Output Characteristics

**Fig2.** Typical  $V_{GS(TH)}$  Gate-Source Voltage Vs.  $T_j$ 

**Fig3.** Typical Transfer Characteristics

**Fig4.** Typical Normalized On-Resistance Vs.  $T_j$ 

**Fig5.** Typical On Resistance Vs Gate-Source Voltage

**Fig6.** Typical On Resistance Vs Drain Current

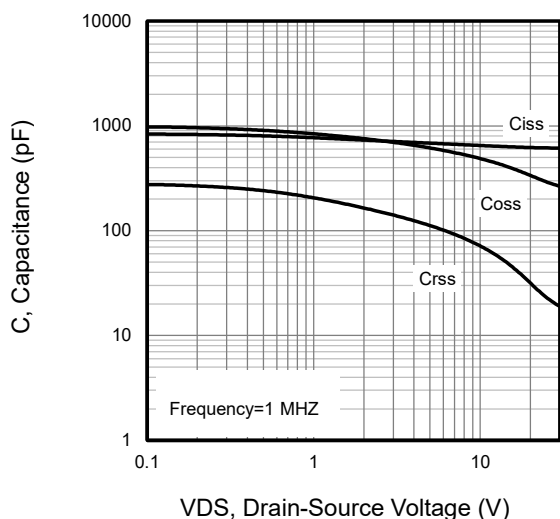
## Q2-Channel Typical Characteristics



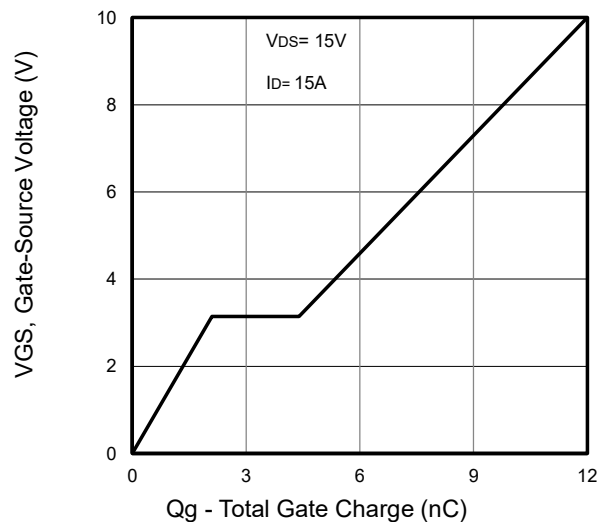
**Fig7.** Typical Source-Drain Diode Forward Voltage



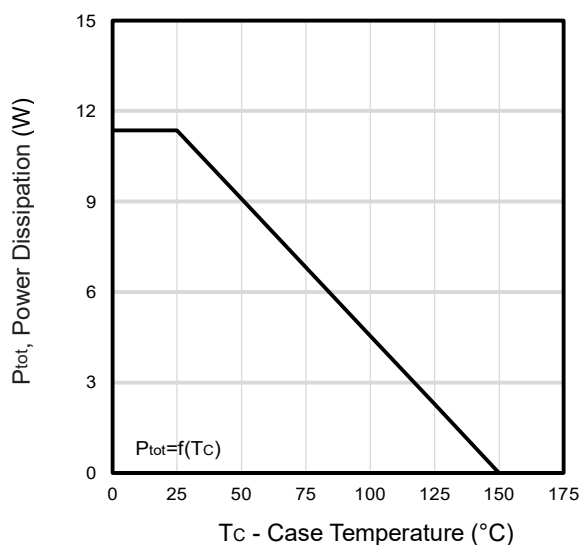
**Fig8.** Maximum Safe Operating Area



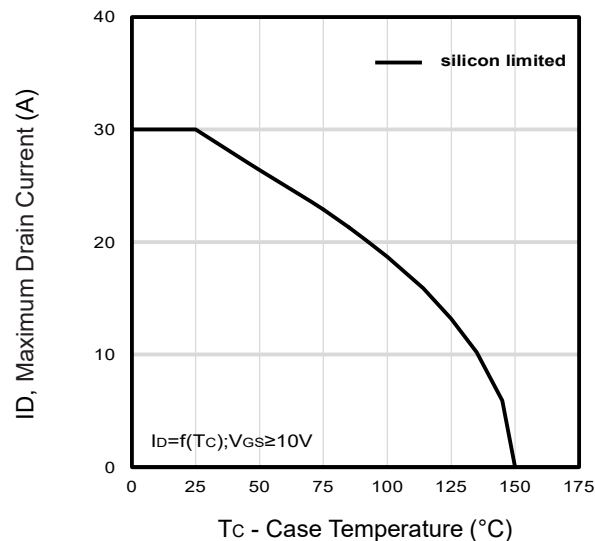
**Fig9.** Typical Capacitance Vs. Drain-Source Voltage



**Fig10.** Typical Gate Charge Vs. Gate-Source Voltage

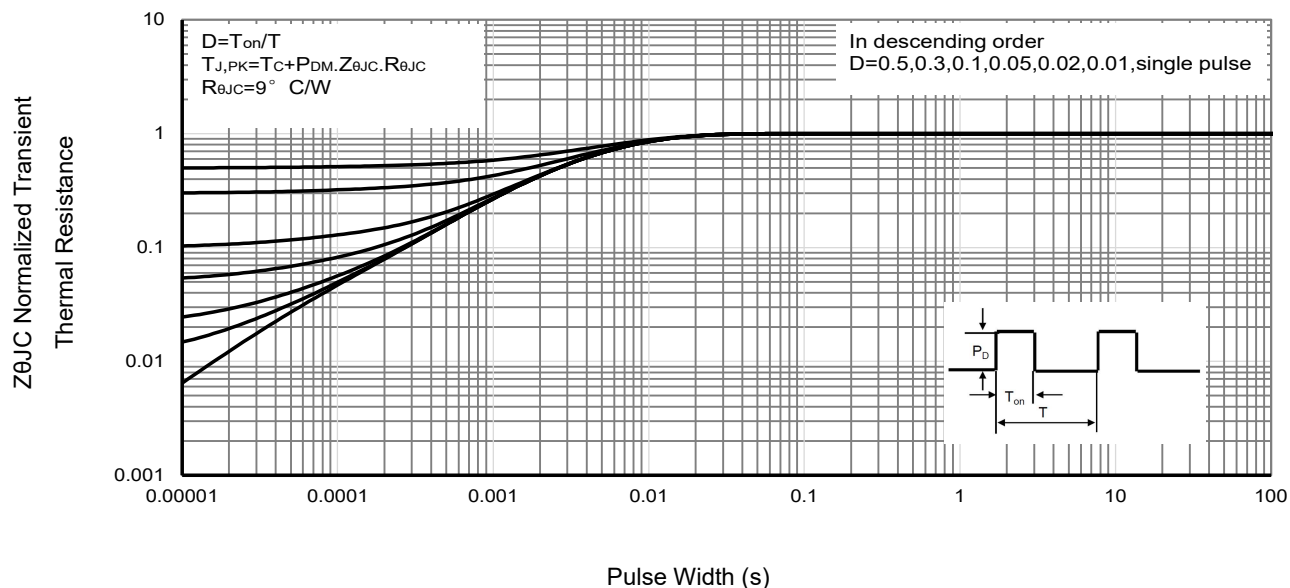


**Fig11.** Power Dissipation Vs. Case Temperature

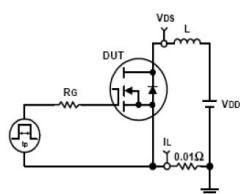


**Fig12.** Maximum Drain Current Vs. Case Temperature

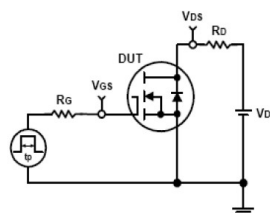
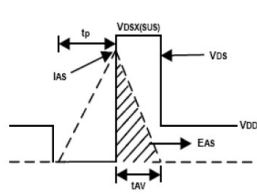
## Q2-Channel Typical Characteristics



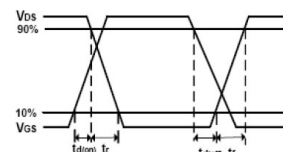
**Fig13 . Normalized Maximum Transient Thermal Impedance**



**Fig14. Unclamped Inductive Test Circuit and waveforms**



**Fig15. Switching Time Test Circuit and waveforms**



**Marking Information**


1<sup>st</sup> line: Vergiga Code (Vs)

2<sup>nd</sup> line: Part Number (3652DB)

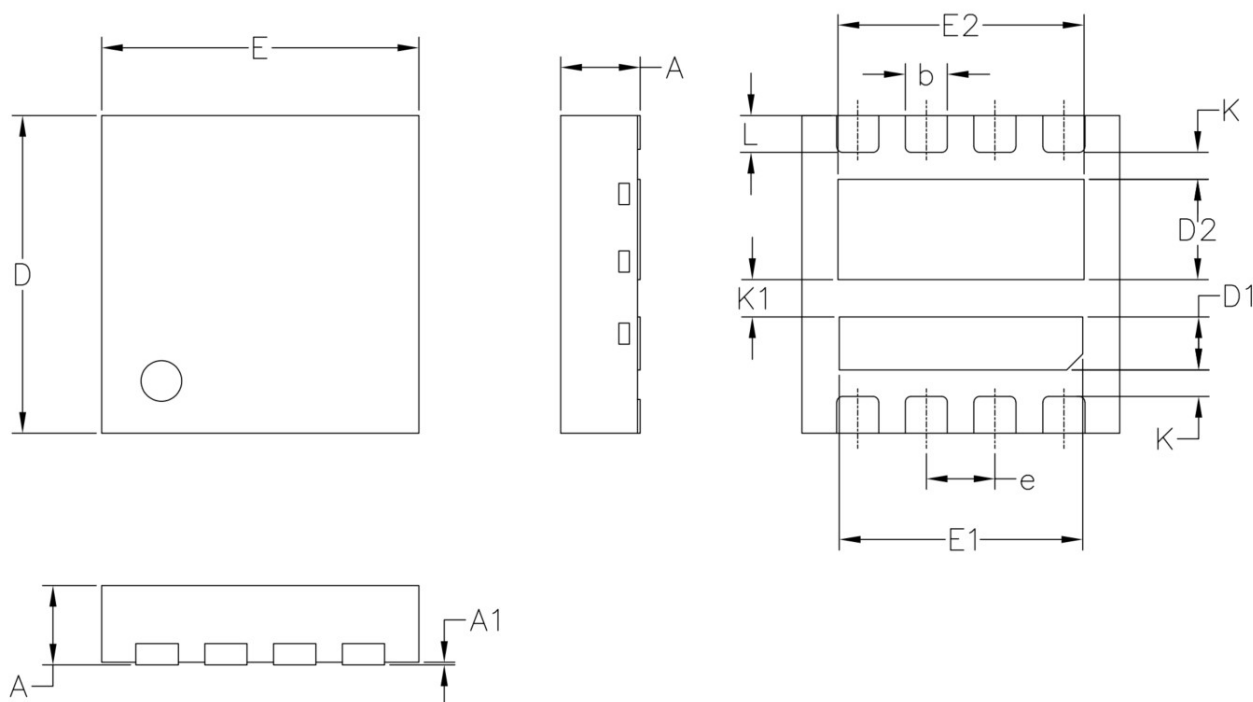
3<sup>rd</sup> line: Date code (XXXYWW)

XXX: Wafer Lot Number Code, code changed with Lot Number

Y: Year Code, refer to table below

WW: Week Code (01 to 53)

| Code | C    | D    | E    | F    | G    | H    | J    | K    | L    | M    | N    | P    | Q    | R    | S    | T    |
|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| Year | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 | 2027 | 2028 | 2029 | 2030 |

**DFN3x3 Package Outline Data**


| Symbol    | Dimensions (unit: mm) |       |      |
|-----------|-----------------------|-------|------|
|           | Min                   | Typ   | Max  |
| <b>A</b>  | 0.70                  | 0.75  | 0.80 |
| <b>A1</b> | 0.00                  | --    | 0.05 |
| <b>b</b>  | 0.35                  | 0.40  | 0.45 |
| <b>D</b>  | 2.90                  | 3.00  | 3.10 |
| <b>D1</b> | 0.40                  | 0.50  | 0.60 |
| <b>D2</b> | 0.85                  | 0.95  | 1.05 |
| <b>E</b>  | 2.90                  | 3.00  | 3.10 |
| <b>E1</b> | 2.20                  | 2.30  | 2.40 |
| <b>E2</b> | 2.20                  | 2.325 | 2.45 |
| <b>e</b>  | 0.55                  | 0.65  | 0.75 |
| <b>K</b>  | 0.15                  | 0.25  | 0.35 |
| <b>K1</b> | 0.25                  | 0.35  | 0.45 |
| <b>L</b>  | 0.27                  | --    | 0.40 |

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